MULTI-DAY WATER TANK SIMULATION

Case Study Report

This case study simulates the water level in a tank over five days. Each day, the program adds incoming water and subtracts outgoing water, checks for overflow or shortage, and displays the final water level.

1. Problem Statement

Develop a Python program to simulate the water level of a tank over five days. The program must update the water level using daily inflow and outflow values, prevent the level from exceeding the tank capacity or falling below zero, display warnings when required, and check whether the threshold level has been reached.

2. Objective

The objective is to understand how variables, lists, loops, conditional statements, and arithmetic operations can be used to model and monitor water levels over multiple days.

3. Assumptions

• The tank has a fixed maximum capacity of 100 units.

• The initial water level is 50 units.

• The threshold water level is 80 units.

• Daily inflow values are [20, 30, 10, 5, 25].

• Daily outflow values are [10, 15, 20, 10, 15].

• The simulation runs for five days, and all water levels are measured in the same units.

4. Algorithm

1. Start the program.

2. Set capacity to 100, initial water level to 50, and threshold to 80.

3. Store the daily inflow and outflow values in lists.

4. Repeat the process for five days.

5. Calculate the new water level by adding inflow and subtracting outflow.

6. If water exceeds capacity, display an overflow warning and set water to capacity.

7. If water falls below zero, display a shortage warning and set water to zero.

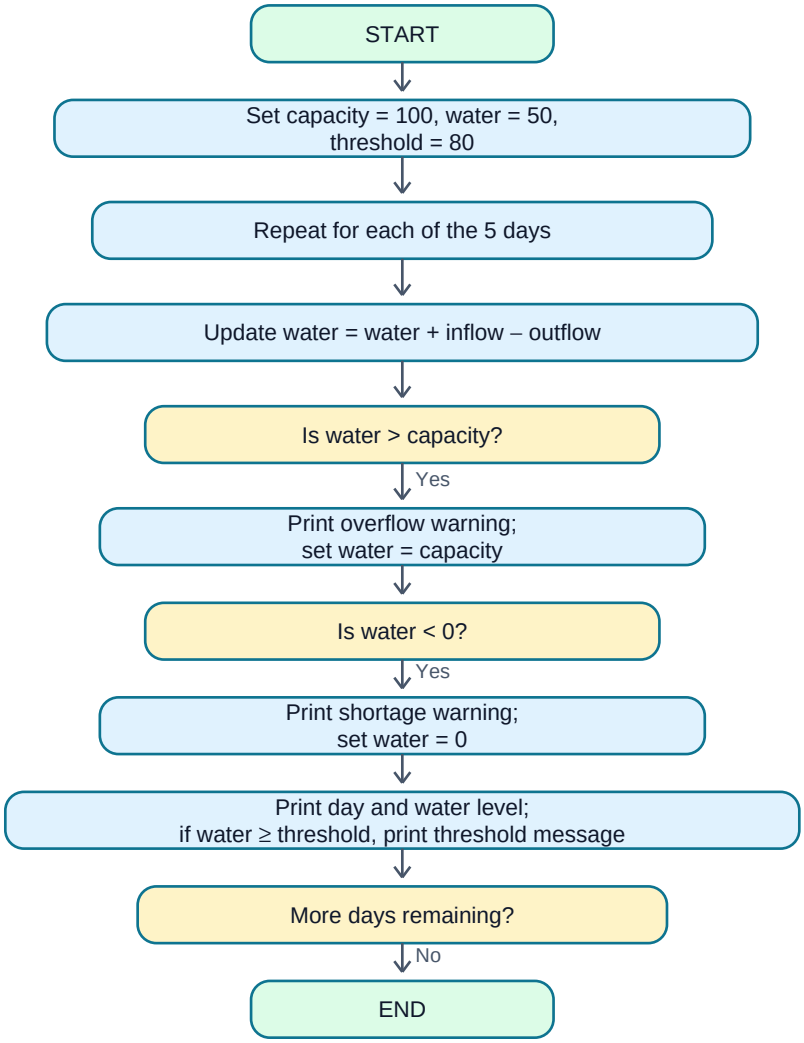
8. Display the day number and current water level.

9. If water is greater than or equal to the threshold, display the threshold message.

10. After all five days, stop the program.

5. Flowchart

The flowchart below shows the main sequence of the simulation. The overflow and shortage checks keep the water level within the valid range.



6. Pseudocode

```
BEGIN
    capacity ← 100
    water ← 50
    threshold ← 80
    inflow ← [20, 30, 10, 5, 25]
    outflow ← [10, 15, 20, 10, 15]

    FOR day ← 0 TO 4
        water ← water + inflow[day] – outflow[day]

        IF water > capacity THEN
            PRINT "Overflow!"
            water ← capacity
        END IF

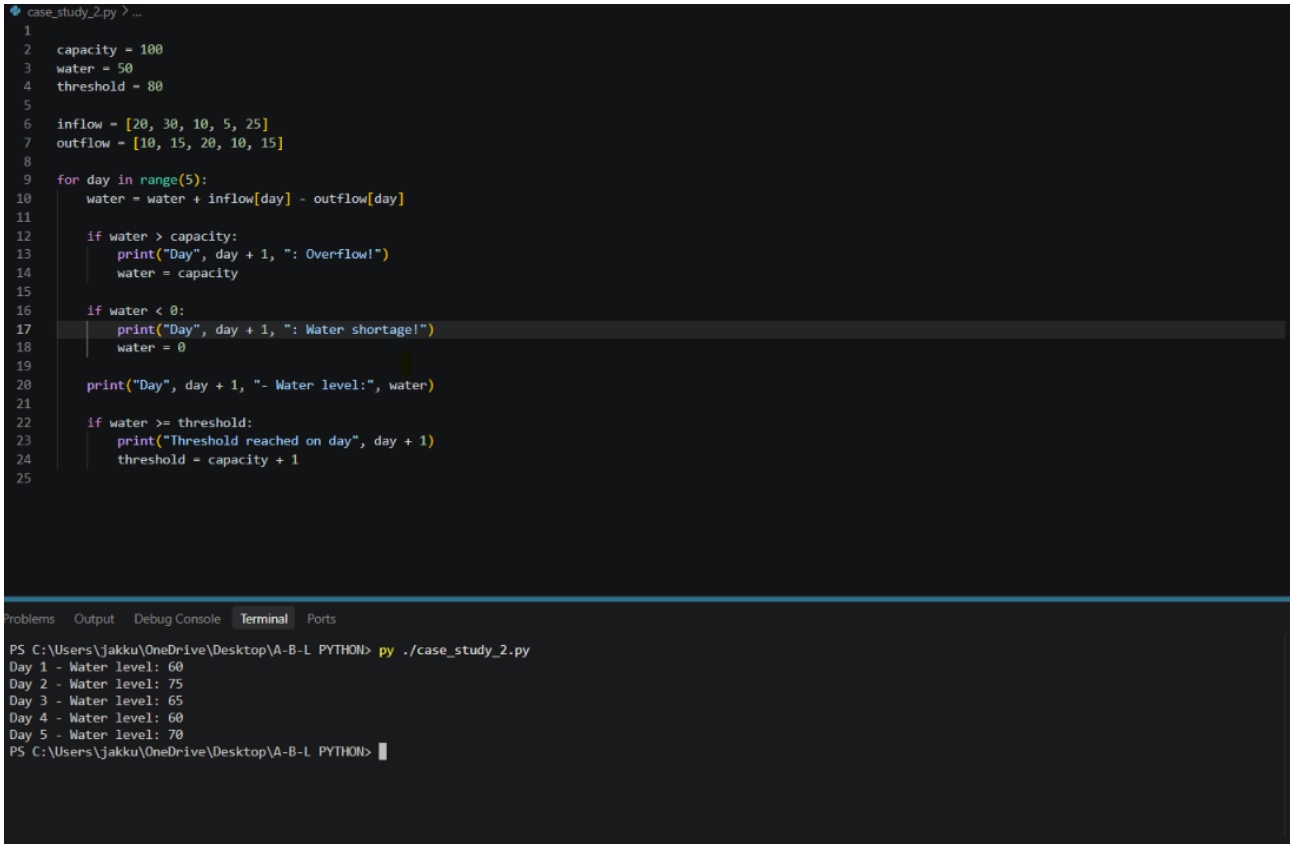
        IF water < 0 THEN
            PRINT "Water shortage!"
            water ← 0
        END IF

        PRINT day + 1, water

        IF water ≥ threshold THEN
            PRINT "Threshold reached"
            threshold ← capacity + 1
        END IF
    END FOR
END
```

7. Screenshot of Python Source Code and Output

The following screenshot contains the Python source code and the terminal output for the simulation.



8. Conclusion

The Multi-Day Water Tank Simulation successfully tracks water levels for five days using daily inflow and outflow data. The output shows water levels of 60, 75, 65, 60, and 70 units. The program demonstrates the use of lists, loops, arithmetic calculations, and conditional statements to monitor a tank and handle possible overflow or shortage conditions.